

Markscheme

Mathematics: analysis and approaches
Practice question bank

114. (a)

Consider the expansion of $(1+x)(1-x)^{-2}$ in ascending powers of x .

Using the expansion of $(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$, find the coefficient of x^3 in $(1+x)(1-x)^{-2}$.

the x^3 term comes from $1 \times 4x^3$ and $x \times 3x^2$ (M1) A1

coefficient $= 4 + 3$ (M1)

$= 7$ A1

[6 marks]